ECON 381: Midterm 2

Professor Muyang Ren

Fall 2026

Name: _____

Instructions

1. Write your name on this exam, your note sheet, and every answer sheet.
2. **Do not open this exam until instructed to do so.**
3. Silence your cell phone and place it in your bag, not in your pocket.
4. This exam is worth 100 points, with 5 additional bonus points available. The maximum recorded score is 100.
5. Check that you have all 7 pages. You have 75 minutes to complete the exam.
6. You may use one 8.5×11 sheet of handwritten notes, on both sides, prepared before the exam. Turn in your note sheet with your exam and answer sheets.
7. Write all answers and supporting work on separate blank sheets of paper. Do not write answers on the exam pages. Clearly number each answer page, label each answer with its section, question number, and part. For each multiple-choice question, write the question number and the corresponding one answer letter.
8. Use other blank sheets for scratch work. Clearly cross out any work you do not want graded in your answer sheets.
9. One standalone, non-programmable calculator is allowed. All other electronic devices are prohibited. For confidence intervals, use the critical values provided in the questions.
10. Sign the honor pledge below. At the end of the exam, turn in this exam, your answer sheets, and your note sheet.
11. **Good luck!**

I pledge on my honor as a University of Tennessee student that I will neither give nor receive aid on this examination.

Signature: _____

1. Multiple Choice Questions (30 points total; 10 questions; 3 points each)

Instructions: On your separate answer sheets, write each question number and **one** answer letter (a, b, c, or d). No explanation is required. Selecting more than one answer earns no credit for that question.

1. In a regression of hourly wage on years of education, which statement expresses least-squares assumption A1 in plain language?

   (a) At every education level, the unobserved wage determinants in the regression error average to zero.
   (b) All workers earn exactly the wage predicted by the population line.
   (c) Every worker has no unobserved determinants of wages.
   (d) The spread of the unobserved wage determinants is the same at every education level.

2. A researcher wants the causal effect of job training on earnings. Let $D_i = 1$ if worker i receives training and 0 otherwise. Which assignment process makes A1 most credible, assuming the regression error is centered at zero?

   (a) Workers enroll when they expect an unobserved improvement in their earnings.
   (b) Workers enroll because their unusually low previous earnings motivates them to receive job training.
   (c) A lottery randomly allocates training, independently of workers' other earnings determinants.
   (d) Employers select workers whose unobserved motivation for working is highest.

3. For a two-sided test of $H_0 : \beta_1 = 0$ in a wage–education regression, what does the p-value measure?

   (a) The fraction of workers whose wages equal their fitted values.
   (b) The probability that the estimated slope equals the true slope.
   (c) The probability that the null hypothesis is true after seeing the data.
   (d) Under the null, the probability of a test statistic at least as far from zero as the observed statistic.

4. Assume A1–A3 and homoskedasticity. If sample size increases from n to $4n$, holding the population error variance and regressor variance fixed, the large-sample variance and standard deviation of the OLS slope become, respectively,

   (a) one-quarter and one-half of their original values.
   (b) unchanged and one-half of their original values.
   (c) one-half and one-quarter of their original values.
   (d) four times and twice their original values.

5. A large-sample regression gives $\hat{\beta}_1 = 61.5$ and $SE(\hat{\beta}_1) = 7.4$. For $H_0 : \beta_1 = 55$ versus $H_1 : \beta_1 \neq 55$, which calculation and 5% decision are correct?

   (a) $t = (61.5 - 55) \times 7.4 = 48.10$; reject.
   (b) $t = (61.5 - 55)/7.4 = 0.88$; do not reject.
   (c) $t = 61.5/7.4 = 8.31$; reject.
   (d) $t = (61.5 - 55)/7.4 = 0.88$; reject.

6. A researcher tests $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ in a test-score–income regression. A Type-I error occurs when the researcher

   (a) reports a positive slope estimate.
   (b) rejects H_0 when the population slope is nonzero.
   (c) does not reject H_0 even though the population slope is zero.
   (d) rejects H_0 even though the population slope is zero.

7. For a two-sided 5% test of $H_0 : \beta_1 = 0$, which set contains the three equivalent rejection rules? Use the same standard error and large-sample normal approximation throughout.

   (a) $|t| > 1.65$; $p < 0.05$; the 99% CI excludes zero.
   (b) $|t| > 1.96$; $p < 0.05$; the 95% CI excludes zero.
   (c) $t > 1.96$; $p < 0.05$; the 90% CI excludes zero.
   (d) $|t| < 1.96$; $p > 0.05$; the 95% CI includes zero.

8. Under A1–A3 and homoskedasticity, compare wage–education populations using the same sample size and measurement units. Which combination makes the large-sample OLS slope more precise?

   (a) Less variation in education with regression-error variance unchanged.
   (b) Larger regression-error variance and less variation in education.
   (c) Smaller regression-error variance and more variation in education.
   (d) Larger regression-error variance with education variation unchanged.

9. Let $D_i = 1$ for districts with $STR_i \geq 20$ and $D_i = 0$ otherwise. Under A1, suppose $E[\text{TestScore}_i \mid D_i = 0] = 668$ and $E[\text{TestScore}_i \mid D_i = 1] = 659$. In the regression $\text{TestScore}_i = \beta_0 + \beta_1 D_i + u_i$,

   (a) $\beta_0 = 668$ and $\beta_1 = -9$.
   (b) $\beta_0 = 659$ and $\beta_1 = 9$.
   (c) $\beta_0 = -9$ and $\beta_1 = 668$.
   (d) $\beta_0 = 668$ and $\beta_1 = 659$.

10. A researcher reruns the same wage–education OLS regression on the same observations, changing only from homoskedasticity-only to heteroskedasticity-robust standard errors. Which statement is correct?

    (a) All residuals must become zero.
    (b) The OLS coefficients stay the same, but their reported standard errors can change.
    (c) Robust standard errors are invalid if the errors are homoskedastic.
    (d) The OLS slope must become smaller.

2. True or False Questions (20 points total; 4 questions; 5 points each)

Instructions: On your separate answer sheets, label each part and write **T** or **F**. You must justify each answer. Full credit requires a correct answer and justification.

(a) If A1–A3 hold and X varies, the interval

$$[\hat{\beta}_1 - 1.96\, SE(\hat{\beta}_1),\ \hat{\beta}_1 + 1.96\, SE(\hat{\beta}_1)]$$

is guaranteed to attain 95% coverage of the population slope when the sample size is $n = 10$.

(b) A researcher finds that OLS residuals average to zero and are uncorrelated with the regressor in the sample. These facts prove A1, $E[u_i \mid X_i] = 0$, for the population model.

(c) The standard error of the estimated wage–education slope estimates the standard deviation of slope estimates across repeated random samples of the same size. The reported standard error itself generally changes from sample to sample.

(d) If wage–education data are heteroskedastic, then the regression errors must have different *means* across different education levels.

3. Analytical Question (50 points total)

Instructions: Answer on your separate answer sheets and label parts (a)–(g), including (g)(i)–(iv). Show your calculations and interpret coefficients using words and units. Use the displayed two-decimal coefficients and standard errors for all calculations; report every calculated answer to two decimal places.

A researcher studies the relationship between education and earnings using `WAGE2.DTA` data on 526 workers. The variable `wage` records average hourly earnings in **dollars per hour**; `educ` records completed **years of education**. Observed education ranges from 0 to 18 years. The researcher estimates

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + u_i$$

with `regress wage educ, robust`. Stata reports the following table, rounded to two decimal places:

```
Linear regression                               Number of obs = 526

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        educ |       0.54       0.06     8.84    0.00         0.42        0.66
       _cons |      -0.90       0.73    -1.25    0.21        -2.33        0.52
------------------------------------------------------------------------------
```

(a) (6 points) Write the sample regression equation, place each coefficient's robust standard error in parentheses below it, and report the sample size.

(b) (6 points) Interpret the education slope. Predict hourly earnings for a high-school graduate with exactly **12 years** of education and a college graduate with exactly **16 years**. Compute the predicted difference, college minus high school.

(c) (4 points) Interpret the intercept. Is the fitted wage at zero years of education economically plausible? Explain, keep in mind of the observed education range.

(d) (6 points) Explain what the two coefficient standard errors estimate. Calculate the t-statistic for testing $H_0 : \beta_1 = 0.40$ against a two-sided alternative. Note that the hypothesized slope is \$0.40 per hour per additional year of education.

Analytical Question (continued)

Use the same wage–education regression. For large-sample normal inference, the two-sided critical values are 1.65 (10%), 1.96 (5%), and 2.58 (1%). The corresponding confidence levels are 90%, 95%, and 99%. Stata computes its reported statistics before display rounding. Recomputed statistics and intervals using the displayed coefficients and SEs can therefore differ slightly; use the rounded values for your calculations.

(e) (6 points) State the null and two-sided alternative for the test in part (d). Use your statistic and the cutoffs to decide at 5% and infer which range contains the two-sided p-value: $p < 0.01$, $0.01 < p < 0.05$, $0.05 < p < 0.10$, or $p > 0.10$. Explain why the displayed Stata t and p columns from the table cannot be used directly for this nonzero null. Does a displayed p-value of `0.00` mean the probability is exactly zero?

(f) (10 points) Compute the 90% and 99% normal-approximation confidence intervals for the education slope. Compare their centers and widths. At the 10%, 5%, and 1% significance levels, is the slope statistically different from **zero**? Use the intervals or cutoffs to justify your decisions.

(g) (12 points total) Evaluate the assumptions for this observational regression. Label your answers (g)(i)–(g)(iv).

(i) (4 points) State A1. Give an omitted determinant of wages that might threaten A1 for a causal education effect, and explain why. Can sample residuals establish A1?

(ii) (3 points) State A2. Discuss what would make it plausible or doubtful for these workers. Can the regression table establish A2?

(iii) (2 points) State A3. What would you examine to assess its plausibility? Can the regression table establish A3?

(iv) (3 points) Explain whether `robust` establishes homoskedasticity, repairs a failure of A1, or resolves dependence between workers.

4. Bonus Question (5 points; 1 point per item)

Instructions: Label parts (a)–(e) on your separate answer sheets. For each item, state whether it **must remain unchanged** or **may change**, and give a brief justification. One point requires both a correct classification and a valid justification.

A researcher switches from `regress testscr str` to `regress testscr str, robust`, using exactly the same observations and the same model with an intercept. The outcome is district test score and the regressor is student–teacher ratio. What happens to each of the following?

(a) The OLS intercept and slope estimates.

(b) R^2.

(c) The standard error of the regression (SER, reported as Root MSE), rather than a coefficient standard error.

(d) The slope's t-statistic for testing a zero slope.

(e) The slope's 95% confidence interval.